

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6

ADOPTADOPTADOPTADOPTSITUATIONALSITUATIONAL

 **ODCS**
BITOL @ LF

 **OpenAPI**
LF

 **AsyncAPI**
LF

 **GraphQL**
LF

 **gRPC**
CNCF

 **OData**
OASIS

DATA PRODUCTS4

ADOPTASSESSASSESSASSESS

 **ODPS**
BITOL @ LF

 **DPDS**
ODM

 **ODPS**
LF

 **DPROD**
OMG

SCHEMA7

ADOPTADOPTADOPTADOPTADOPTASSESS

 **XML Schema**
W3C

 **JSON Schema**
IETF

 **SQL DDL**
ISO/IEC

 **AVRO Schema**
ASF

 **Protobuf**
GOOGLE

 **LinkML**
LINKML

ASSESS

 **Table Schema**
FRICITIONLESS

SEMANTICS8

SITUATIONALSITUATIONALSITUATIONALSITUATIONALSITUATIONASSESS

 **RDF/OWL**
W3C

 **DCAT**
W3C

 **SKOS**
W3C

 **SHACL**
W3C

 **JSON-LD**
W3C

 **Ossie**
ASF

ASSESSASSESS

 **schema.org**
W3C

 **ShEx**
W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS8

ADOPTADOPTADOPTADOPTADOPTADOPTSITUATIONASSESS

 **CSV**
IETF

 **JSON**
IETF

 **XML**
W3C

 **YAML**
YAML.ORG

 **PARQUET**
ASF

 **AVRO**
ASF

 **ORC**
ASF

 **Lance**
LANCE

OPEN TABLE FORMATS4

ADOPTSITUATIONASSESSASSESS

 **Iceberg**
ASF

 **Delta**
LF

 **Hudi**
ASF

 **Lance**
LANCE

STORAGE SYSTEMS2

ADOPTCAUTION

 **S3**
AWS

 **HDFS**
ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY5

ADOPTADOPTADOPTASSESSCAUTION

 **JDBC**
JCP

 **ODBC**
ISO/IEC

 **ADBC**
ASF

 **Quack**
DUCKDB LABS

 **XMLA**
MICROSOFT

INTERCONNECTION3

ADOPTSITUATIONCAUTION

 **HTTP**
IETF

 **Delta Sharing**
LF

 **FTP / SFTP**
IETF

MESSAGING5

ADOPTADOPTSITUATIONALSITUATIONCAUTION

 **Kafka**
ASF

 **CloudEvent**
s CNCF

 **MQTT**
OASIS

 **AMQP**
OASIS

 **JMS**
JAKARTA EE

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2

ADOPTADOPT

 **DataFrame API**
DATA-APIs

 **Apache Arrow**
ASF

PROCESSING7

ADOPTADOPTADOPTADOPTSITUATIONASSESSCAUTION

 **Spark**
ASF

 **Pandas**
NUMFOCUS

 **SQL DML**
ISO/IEC

 **dbt**
DBT LABS

 **Beam**
ASF

 **Ibis**
IBIS

 **XSLT**
W3C

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS5

ADOPTADOPTSITUATIONASSESSCAUTION

 **Iceberg Catalog**
ASF

 **Schema Registry**
CONFLUENT

 **Unity Catalog**
LF

 **DuckLake**
DUCKDB LABS

 **Hive Metastore**
ASF

LINEAGE2

ADOPTASSESS

 **OpenLineage**
e LF

 **PROV**
W3C

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY5

ADOPTSITUATIONALSITUATIONALSITUATIONCAUTION

 **SQL**
ISO/IEC

 **Substrait**
LF

 **SPARQL**
W3C

 **GQL**
ISO/IEC

 **MDX**
MICROSOFT

DATA QUALITY3

SITUATIONALSITUATIONALSITUATION

 **Great Expectations**
GX LABS

 **dbt tests**
DBT LABS

 **SodaCL**
SODA

OBSERVABILITY2

ADOPTASSESS

 **OpenTelemetry**
CNCF

 **OORS**
BITOL @ LF

POLICIES2

ADOPTASSESS

 **OPA**
CNCF

 **ODRL**
W3C

AI INTERFACES2

ADOPTSITUATION

 **MCP**
LF

 **A2A**
LF